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PD90

SEAT No. :
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[6410]-411

**T.E. (Chemical) (Insem)
 MASS TRANSFER - II**

(2019 Pattern) (Semester - II) (309349)

Time : 1 Hour]

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data if necessary.

[Max. Marks : 30

Q1) a)

A liquid mixture containing 40 mole% n-heptane and 60 mole% n-octane is subjected to differential distillation at atmospheric pressure, with 60 mole% of the liquid distilled. Find the composition of the composited distillate and the residue.

[10]

Data : Relative volatility for the given system is 2.16.

b) Explain Positive Deviation from ideality with neat sketch.

[5]

OR

Q2) a)

A feed containing 60 mole % hexane and 40 mole% octane is fed to a pipe still through a pressure reducing valve into a flash separator. The vapour and liquid leaving the separator is assumed to be in equilibrium. If 50mole% of the feed is vaporized. Find the composition of the top and bottom products. The equilibrium data is given below.

[10]

x, mole fraction of hexane in liquid	1.0	0.69	0.40	0.192	0.045	0
y, mole fraction of hexane in vapour	1.0	0.932	0.78	0.538	0.1775	0

b) Derive the Rayleigh's equation.

[5]

Q3) a)

A mixture of benzene and toluene containing 60 mole% benzene is to be separated to give a product of 95 mole% benzene and a bottom product containing 10 mole% benzene. The feed enters a column at its bubble point. It is proposed to operate the column with reflux ratio of 2.5. It is required to find the number of theoretical plates needed and the position of feed plate.

[10]

Equilibrium data:

x	0	0.05	0.21	0.27	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
y	0	0.13	0.21	0.375	0.5	0.6	0.7	0.77	0.83	0.9	0.95	1.0

b) Derive Fenske Underwood equation to calculate minimum number of plates at total reflux.

[5]

OR

Q4)

A continuous rectifying column handles a mixture consisting of 40 per cent of benzene by mass and 60 per cent of toluene at the rate of 4kg/s, and separates it into a product containing 97 per cent of benzene and a liquid containing 98 per cent toluene. The feed is liquid at its boiling-point.

[15]

a) Calculate the mass flows of distillate and waste liquor.

b) If a reflux ratio of 3.5 is employed, how many plates are required in the rectifying part of the column?

c) What is the actual number of plates if the plate-efficiency is 60 Per cent?

x, mole fraction of benzene in liquid	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
y, mole fraction of benzene in vapour	0.22	0.38	0.51	0.63	0.7	0.78	0.85	0.91	0.96
